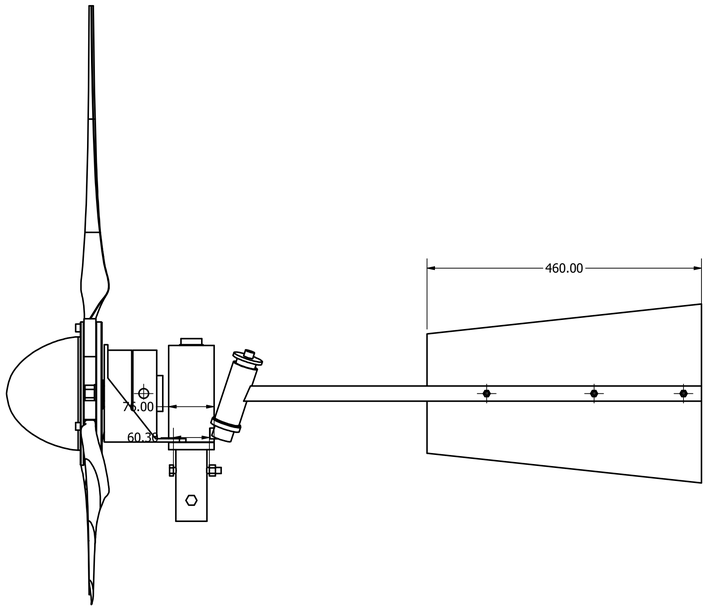
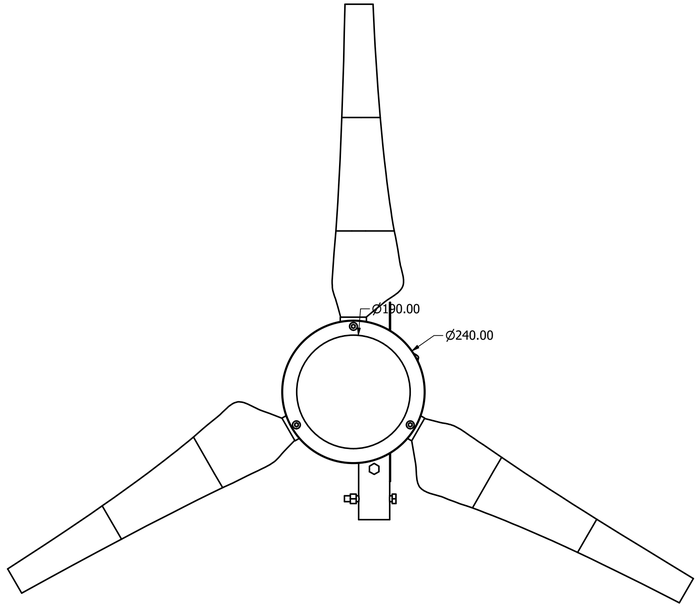
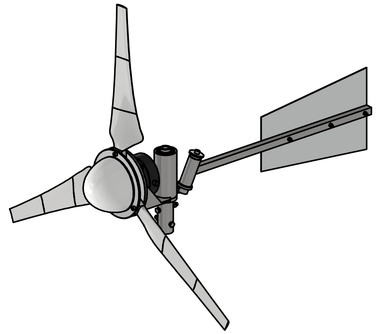


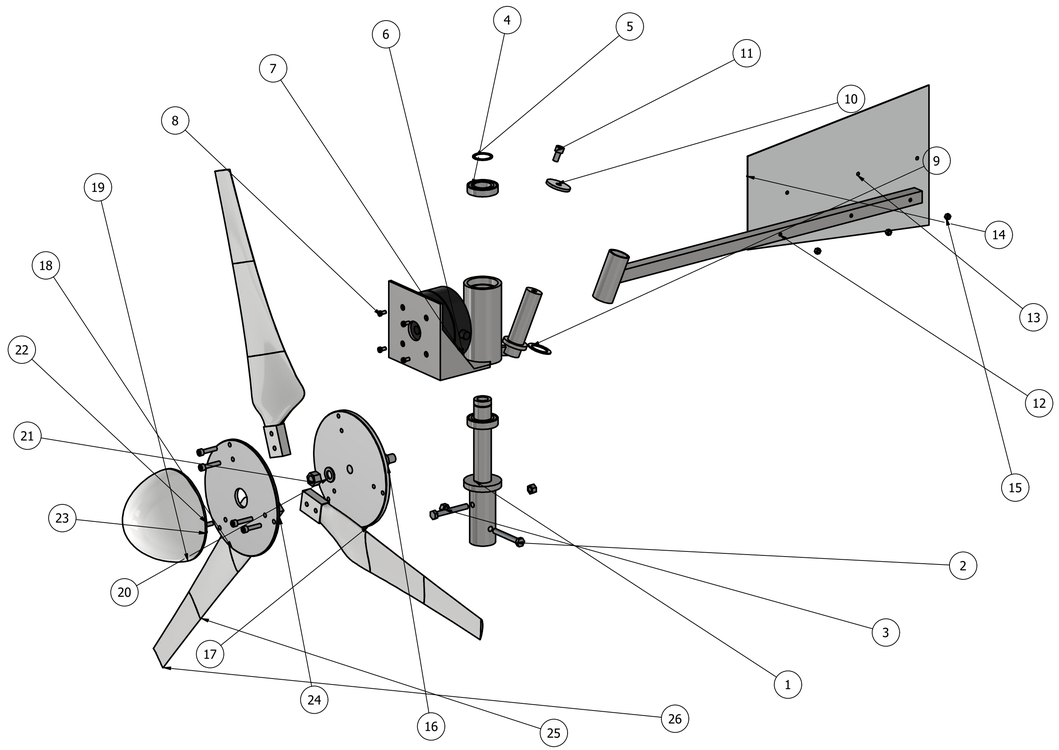



WINDTURBINE - 1.3 M 3-BLADE UPWIND SMALL WIND TURBINE (DESIGNED AND CALCULATED, NOT BUILT)
ROTOR D 1300, SG6040 -> SG6043 BLADES, SCHMITZ CHORD 125 -> 47 / TWIST 20.9 -> 4.0 DEG FOR TSR 5; ASA PRINTED IN 3 SEGMENTS, CF ROD JOINTS
GENERATOR ATO-PMG-NE400WM 24 V 3-PHASE PM, DIRECT DRIVE; HUB HEIGHT 6215, GUY COLLAR 4600, 2IN SCH 40 MAST, TILT-UP ON A HINGED BASE
CALC.: CP MAX 0.42 AT TSR 4.25; 294 W AT 11 M/S (704 RPM), MAX 448 W; CHARGING FROM 3.6 M/S
FURLING TAIL: HINGE 20 DEG, ROTOR OFFSET 35; FURL STARTS AT 8.5 M/S (40 N THRUST), 54 DEG AT 20 M/S
LOADS IEC 61400-2 CLASS IV: PARKED 42 M/S HEAD FORCE 268 N, MAST 57 MPA (SF 4.2); BLADE ROOT SF 1.9
ASSEMBLY PARAMETERS: Yaw (NACELLE), Spin (ROTOR), Furl (TAIL, 0-80 DEG).

| | | |
|---|---|---|
| DRAWN Jay Patel | 10/29/2026 | |
| CHECKED | | TITLE WindTurbine - general assembly |
| QA | | |
| MFG | | |
| APPROVED | | SIZE D · DWG NO WT-DWG-001 · REV |
| | | SCALE 1 / 50 · SHEET 1 OF 2 |




| PARTS LIST | | | |
|---|---|---|---|
| ITEM | QTY | PART NUMBER | DESCRIPTION |
| 1 | 1 | WT-402 | Yaw spindle 1045: 52.2 plug in the 2in pipe (2 x M10 cross bolts), 76 collar on the pipe end, 35 h6 spindle for 2 x 6007-2RS, DIN 471 groove, 22 bore for the generator cable |
| 2 | 2 | WT-516 | Hex bolt ISO 4014 M10 x 80 8.8 zinc (yaw spindle and guy collar cross bolts) |
| 3 | 2 | WT-519 | Nylon-insert lock nut DIN 985 M10 8 zinc |
| 4 | 2 | WT-403 | Deep groove ball bearing 6007-2RS 35 x 62 x 14, sealed (C 16.8 kN, C0 10.2 kN, SKF data) |
| 5 | 1 | WT-511 | Retaining ring DIN 471 35 x 1.5 (yaw spindle, above the upper 6007) |
| 6 | 1 | WT-401 | Nacelle frame, welded S235: yaw tube 76 (2 x 6007 seats 62 K6, bored after welding), 6 mm generator plate (40 hole, 4 x M6 C'bore on PCD 110), 6 mm arm plate, 2 gussets, hinge bracket and 33.4 hinge post tilted 20 deg (1in Sch 80 pipe, M10 tapped end); zinc-rich primer + paint |
| 7 | 1 | WT-301 | Permanent-magnet generator ATO-PMG-NE400WM, 24 V 3-phase AC, 400 W at 800 rpm (vendor curve), max 445 W, start torque <= 0.35 Nm, 4.5 kg, IP54, class F; envelope from the vendor drawing (145 x 82, M16 shaft); mounting PCD 110 assumed |
| 8 | 4 | WT-514 | Low-head cap screw DIN 7984 M6 x 12 8.8 (generator to the frame plate, C'bored flush; Loctite 243) |
| 9 | 1 | WT-407 | PTFE thrust washer 48 x 34.4 x 1.9 (tail sleeve on the hinge collar) |
| 10 | 1 | WT-406 | Hinge cap washer S235 50 x 5 (keeps the tail sleeve on the post), M10 x 20 into the post end |
| 11 | 1 | WT-515 | SHCS ISO 4762 M10 x 20 8.8 (hinge cap into the post) |
| 12 | 1 | WT-404 | Tail boom, welded S235: 1-1/4in Sch 40 sleeve (35.0 bore on the 33.4 post, 0.8 clearance) + 1in x 1in x 2 square tube 780 long, welded at 90 + 20 deg so the boom is level at rest; 3 x 6.4 for the vane bolts |
| 13 | 1 | WT-405 | Tail vane 5052-H32 1.6 mm, 460 long, 200 -> 300 high (0.115 m2), 3 x M6 to the boom |
| 14 | 3 | WT-518 | Hex bolt ISO 4014 M6 x 40 A2 (tail vane to boom) |
| 15 | 3 | WT-521 | Nylon-insert lock nut DIN 985 M6 A2 (tail vane) |
| 16 | 1 | WT-301S | Generator rotor shaft (part of the ATO-PMG-NE400WM, split out so the rotor can turn in the model): 17 mm, 25 collar, M16x1.5 thread |
| 17 | 1 | WT-201 | Hub disc 6061-T6: 240 x 10 with 45 spigot, bore 17 H7 on the generator shaft (clamped by the M16x1.5 nut against the shaft collar), 6 x M8 tapped through on r 80 / 110 for the blade roots |
| 18 | 1 | WT-202 | Blade clamp plate 6061-T6 240 x 6, 40 centre hole (nut access), 6 x 8.6 for the blade bolts (the inner 3 also clamp the spinner flange) |
| 19 | 1 | WT-203 | Spinner (nose cone) 190 x 120, 2.4 wall, PETG printed flange down; flange clamped under the 3 inner M8 blade bolts |
| 20 | 1 | WT-523 | Washer ISO 7089 17 x 30 x 2.5 A2 (hub nut) |
| 21 | 1 | WT-522 | Nylon-insert lock nut DIN 985 M16 x 1.5 fine A2 (hub on the generator shaft, 60 Nm: clamps the hub against the shaft collar) |
| 22 | 3 | WT-513 | SHCS ISO 4762 M8 x 40 A2-70 (inner blade bolts, also clamp the spinner flange; Loctite 243, 12 Nm) |
| 23 | 3 | WT-512 | SHCS ISO 4762 M8 x 35 A2-70 (clamp plate + blade root into the hub thread; Loctite 243, 12 Nm) |
| 24 | 3 | WT-101 | Blade root segment r 62-270: root block 44 x 20 (2 x 8.6 holes for M8 on r 80 / 110), SG6040 section, chord 125, twist 20.9 deg; 2 x 5.2 rod holes; printed ASA, span vertical |
| 25 | 3 | WT-102 | Blade mid segment r 270-460: SG6040 -> SG6043 blend, chord 100 -> 65, twist 13.6 -> 7.0 deg; 2 x 5.2 + 2 x 3.2 rod holes; printed ASA, span vertical |
| 26 | 3 | WT-103 | Blade tip segment r 460-650: SG6043, chord 65 -> 47, twist 7.0 -> 4.0 deg; 2 x 3.2 rod holes; printed ASA, span vertical |
| 27 | 6 | WT-104 | Pultruded carbon fibre rod 5 mm x 130 mm (cut from 1 m), bonded across a blade joint with epoxy |
| 28 | 6 | WT-105 | Pultruded carbon fibre rod 3 mm x 90 mm (cut from 1 m), bonded across a blade joint with epoxy |

TOWER PARTS (NOT IN THIS VIEW, SEE SHEET 1 AND THE BOM): WT-501 / 502 MAST 2IN SCH 40 x 10 FT, WT-503 COUPLING, WT-504 GUY COLLAR, WT-505 HINGED BASE, WT-516 / 517 / 519 / 520 / 524 BOLTS AND NUTS. GUYS, ANCHORS, ELECTRONICS AND BATTERIES: BOM 'NOT MODELLED' SECTION.

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|---|---|---|
| DRAWN Jay Patel | 10/29/2026 | |
| CHECKED | | TITLE: WindTurbine - general assembly |
| QA | | |
| MFG | | |
| APPROVED | | |
| | | SIZE D · DWG NO WT-DWG-001 · REV |
| | | SCALE 1 / 6 · SHEET 2 OF 2 |